

CS376 – HW4 Report

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[Task 1]

Goal: Derive gradient of neural ODE.

Assume a neural ODE $\frac{dz(t)}{dt} = f(z(t), t, \theta)$ where f is a neural network, and $z(t_0)$ is given. We can obtain $z(t_1) = z(t_0) + \int_{t_0}^{t_1} f(z(t), t, \theta) dt$ by integrating the ODE from t_0 to t_1 . The loss function $L = L(z(t_1))$ depends on the final state $z(t_1)$. We can derive gradient $\frac{\partial L}{\partial \theta}$ using adjoint sensitivity method. Define adjoint state, $a(t) = \frac{\partial L}{\partial z(t)}$. From $a(t_1) = \frac{\partial L}{\partial z(t_1)}$, we can compute $a(t_0)$ by differentiating it with respect to t , $\frac{da(t)}{dt} = -a(t)^T \frac{\partial f(z(t), t, \theta)}{\partial z(t)}$, and by integrating it from t_1 to t_0 . Then, we obtain $a(t_0) = a(t_1) + \int_{t_1}^{t_0} \frac{da(t)}{dt} dt = a(t_1) - \int_{t_1}^{t_0} a(t)^T \frac{\partial f(z(t), t, \theta)}{\partial z(t)} dt$. Finally, apply the chain rule to derive the gradient of the loss with respect to θ : $\frac{\partial L}{\partial \theta} = - \int_{t_1}^{t_0} a(t)^T \frac{\partial f(z(t), t, \theta)}{\partial \theta} dt = \int_{t_0}^{t_1} a(t)^T \frac{\partial f(z(t), t, \theta)}{\partial \theta} dt$.

[Task 2]

Goal: Derive the denoising score matching loss.

In the original score matching, we have to minimize the loss $\mathcal{L}_{\text{SM}} = \mathbb{E}_{p(x)}[\|\nabla_x \log p(x) - s_\theta(x)\|^2]$ where $p(x)$ is the data distribution and $s_\theta(x)$ is the score function of the model. Then, $\mathcal{L}_{\text{SM}} = \mathbb{E}_{p(x)}[\|s_\theta(x)\|^2 - 2s_\theta(x)^T \nabla_x \log p(x) + \|\nabla_x \log p(x)\|^2] = \mathbb{E}_{p(x)}[\|s_\theta(x)\|^2 + 2\nabla_x \cdot s_\theta(x)] + C$ because the last term $\|\nabla_x \log p(x)\|^2$ is a constant. However, this formulation has limitations that it can be unstable to noisy samples and hard to evaluate because $p(x)$ becomes very small in low-density regions where data are sparse.

To resolve this, we introduce perturbed data with Gaussian noise, $\tilde{x} \sim p_\sigma(\tilde{x}|x) = \mathcal{N}(\tilde{x}; x, \sigma^2 I)$. Using this, we can derive the denoising score matching loss as: $\mathcal{L}_{\text{DSM}} = \mathbb{E}_{p(x)} \mathbb{E}_{p_\sigma(\tilde{x}|x)} [\|\nabla_{\tilde{x}} \log p_\sigma(\tilde{x}|x) - s_\theta(\tilde{x})\|^2]$. Let $p(x, \tilde{x}) = p(x)p_\sigma(\tilde{x}|x)$, then $\mathcal{L}_{\text{DSM}} = \mathbb{E}_{p(x, \tilde{x})} [\|s_\theta(\tilde{x})\|^2 - 2s_\theta(\tilde{x})^T \nabla_{\tilde{x}} \log p_\sigma(\tilde{x}|x) + \|\nabla_{\tilde{x}} \log p_\sigma(\tilde{x}|x)\|^2] = \mathbb{E}_{p(x, \tilde{x})} [\|s_\theta(\tilde{x})\|^2 - 2s_\theta(\tilde{x})^T \nabla_{\tilde{x}} \log p_\sigma(\tilde{x}|x)] + C$. In specific condition, $x \approx \tilde{x}$ so that $\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{DSM}} + C$. Those two losses have the same optimal parameters θ^* . Thus, when we learn score-based model, the denoising score matching loss can be used.